Object Detection II

CS 4391 Introduction Computer Vision

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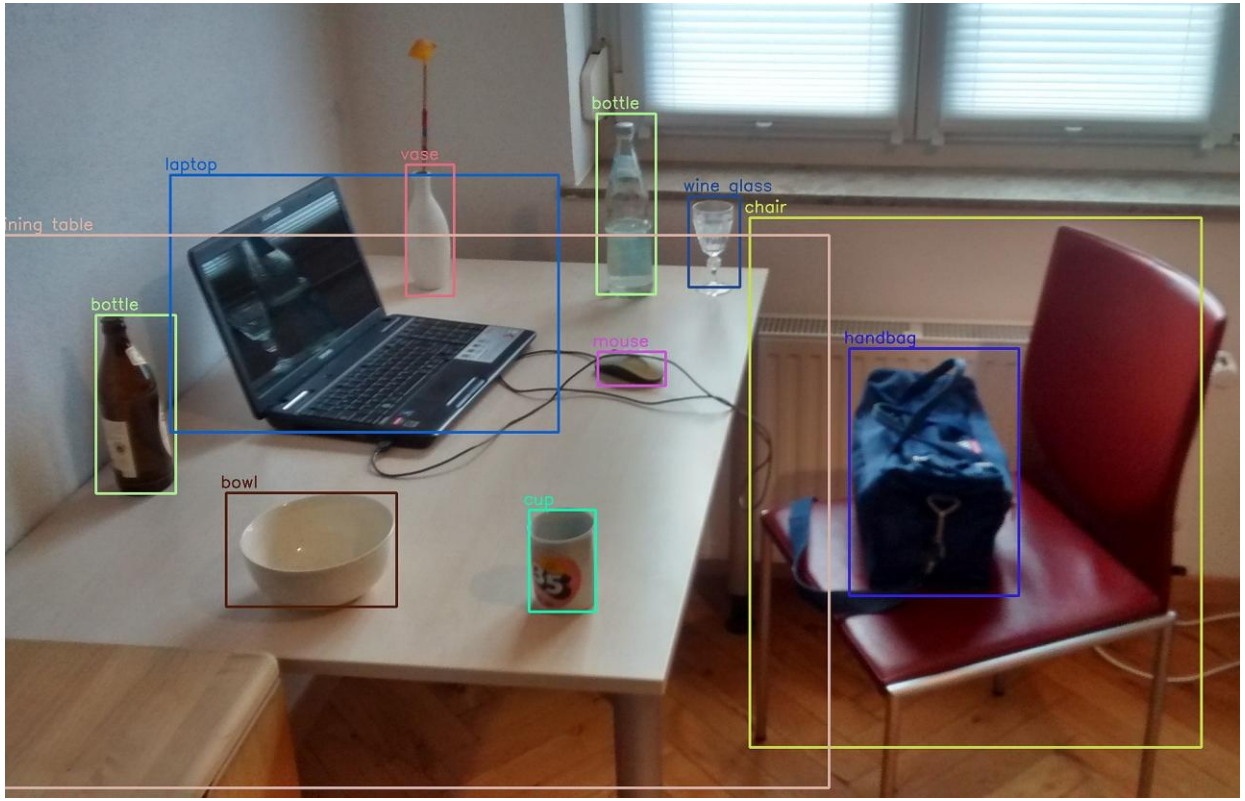
Object Detection

- Localize objects in images and classify them



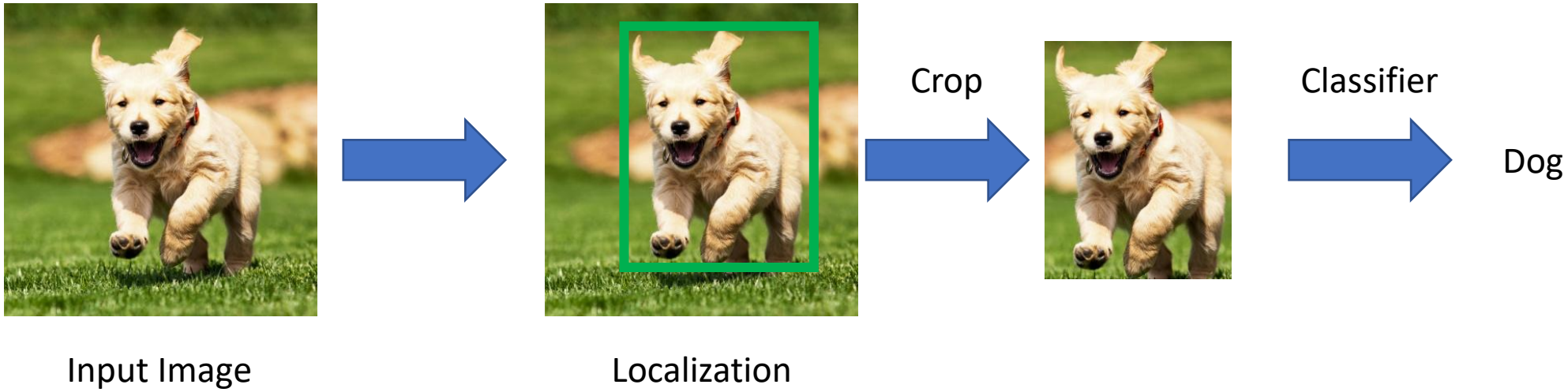
Wikipedia

Why using bounding boxes?

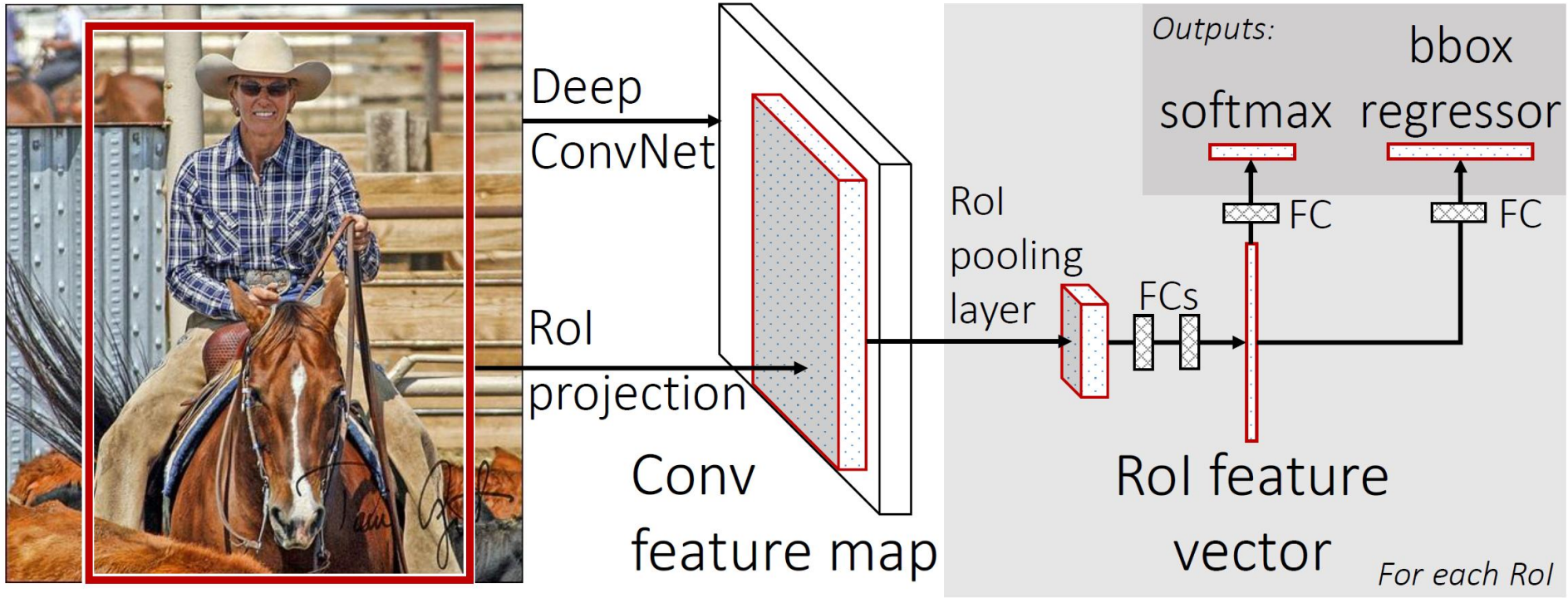
- Easy to store
  - (x, y, w, h): box center with width, height
  - (x1, y1, x2, y2): top left corner and bottom right corner
- Easy for image processing
  - Crop a region

Object Detection

- Localization + Classification



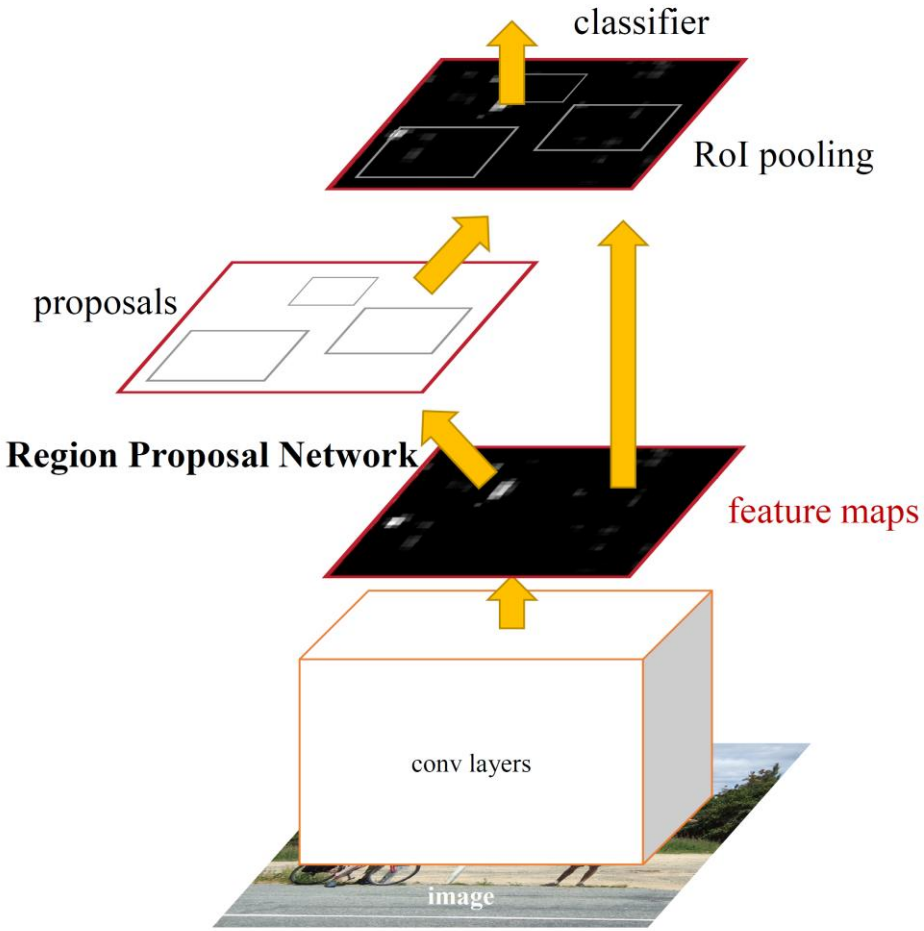
Fast R-CNN



Fast R-CNN. Girshick, ICCV, 2015

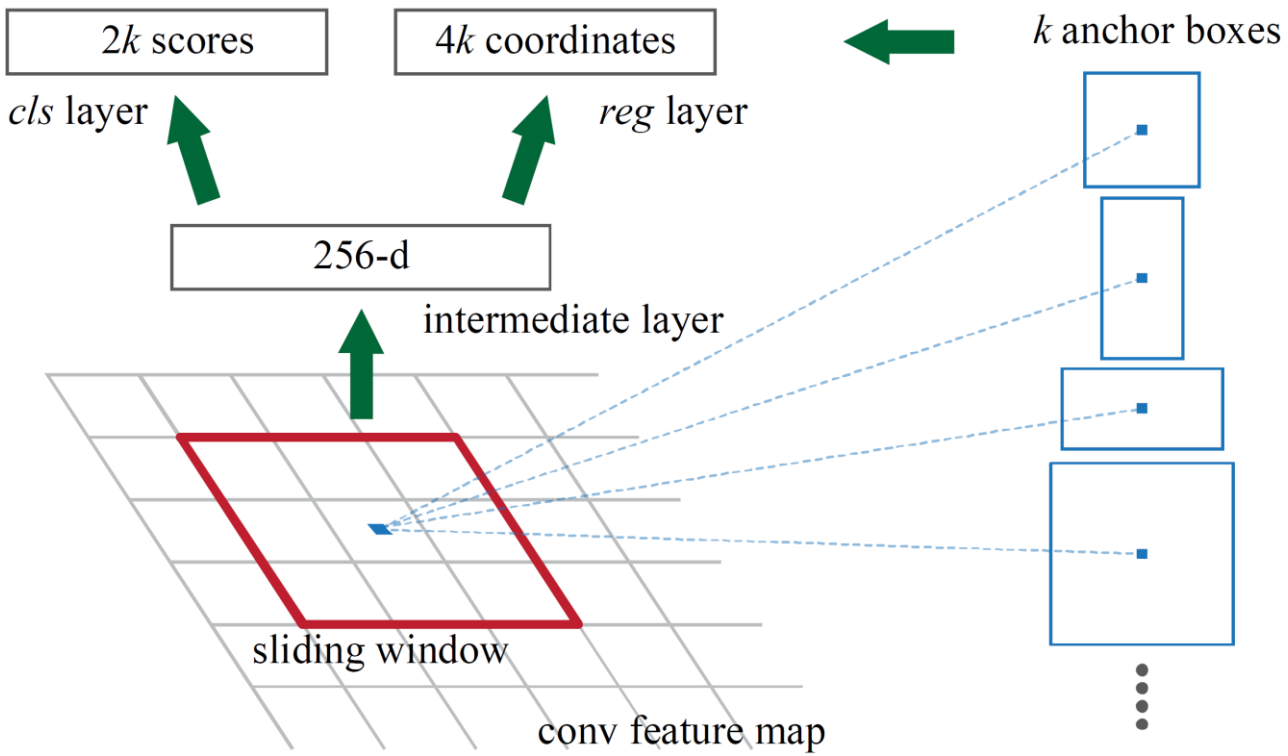
Faster R-CNN

- A single network for object detection
  - Region proposal network
  - Classification network



Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks. Ren et al., NeurIPS, 2015

Region Proposal Network



An anchor is centered at the sliding window in question, and is associated with a scale and aspect ratio (3 scales, 3 aspect ratios, k=9)

3x3 conv layer to 256-d

```
layer {
  name: "rpn_conv/3x3"
  type: "Convolution"
  bottom: "conv5"
  top: "rpn/output"
  param { lr_mult: 1.0 }
  param { lr_mult: 2.0 }
  convolution_param {
    num_output: 256
    kernel_size: 3 pad: 1 stride: 1
    weight_filler { type: "gaussian" std: 0.01 }
    bias_filler { type: "constant" value: 0 }
  }
}
```

classification

```
layer {
  name: "rpn_cls_score"
  type: "Convolution"
  bottom: "rpn/output"
  top: "rpn_cls_score"
  param { lr_mult: 1.0 }
  param { lr_mult: 2.0 }
  convolution_param {
    num_output: 18   # 2(bg/fg) * 9(anchors)
    kernel_size: 1 pad: 0 stride: 1
    weight_filler { type: "gaussian" std: 0.01 }
    bias_filler { type: "constant" value: 0 }
  }
}
```

Bounding box regression

```
layer {
  name: "rpn_bbox_pred"
  type: "Convolution"
  bottom: "rpn/output"
  top: "rpn_bbox_pred"
  param { lr_mult: 1.0 }
  param { lr_mult: 2.0 }
  convolution_param {
    num_output: 36   # 4 * 9(anchors)
    kernel_size: 1 pad: 0 stride: 1
    weight_filler { type: "gaussian" std: 0.01 }
    bias_filler { type: "constant" value: 0 }
  }
}
```

Faster R-CNN

Table 1: Detection results on **PASCAL VOC 2007 test set** (trained on VOC 2007 trainval). The detectors are Fast R-CNN with ZF, but using various proposal methods for training and testing.

train-time region proposals		**test-time region proposals**		
method	# boxes	method	# proposals	mAP (%)
SS	2k	SS	2k	58.7
EB	2k	EB	2k	58.6
RPN+ZF, shared	2k	RPN+ZF, shared	300	**59.9**

ZF: M. D. Zeiler and R. Fergus. Visualizing and understanding convolutional neural networks. In ECCV, 2014

Table 4: **Timing** (ms) on a K40 GPU, except SS proposal is evaluated in a CPU. "Region-wise" includes NMS, pooling, fc, and softmax. See our released code for the profiling of running time.

model	system	conv	proposal	region-wise	total	rate
VGG	SS + Fast R-CNN	146	1510	174	1830	0.5 fps
VGG	RPN + Fast R-CNN	141	**10**	47	**198**	**5 fps**
ZF	RPN + Fast R-CNN	31	**3**	25	**59**	**17 fps**

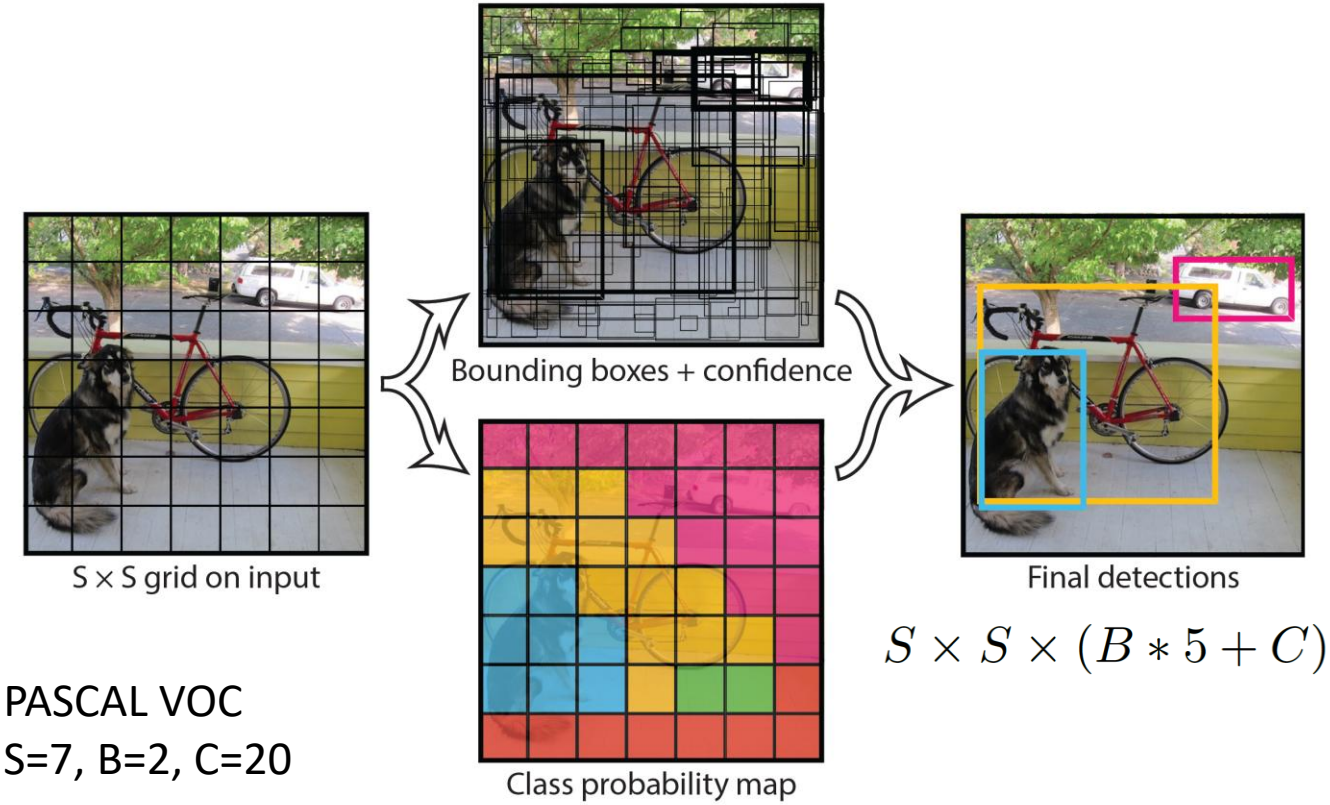
Two stage vs One stage

- Two stage detection methods
  - Stage 1: generate region proposals
  - Stage 2: classify region proposals and refine their locations
  - E.g., R-CNN, Fast R-CNN, Faster R-CNN

- One stage detection methods
  - An end-to-end network for object detection
  - E.g., YOLO

YOLO

- Regress to bounding box locations and class probabilities



$$S \times S \times (B * 5 + C)$$

PASCAL VOC
S=7, B=2, C=20

- Each grid handles objects with centers (x, y) in it
- Each grid predicts B bounding boxes
- Each bounding box predicts (x, y, w, h) and confidence (IoU of box and ground truth box)

$$\Pr(\text{Object}) * \text{IOU}_{\text{pred}}^{\text{truth}}$$

- Each grid also predicts C class probabilities

$$\Pr(\text{Class}_i | \text{Object})$$

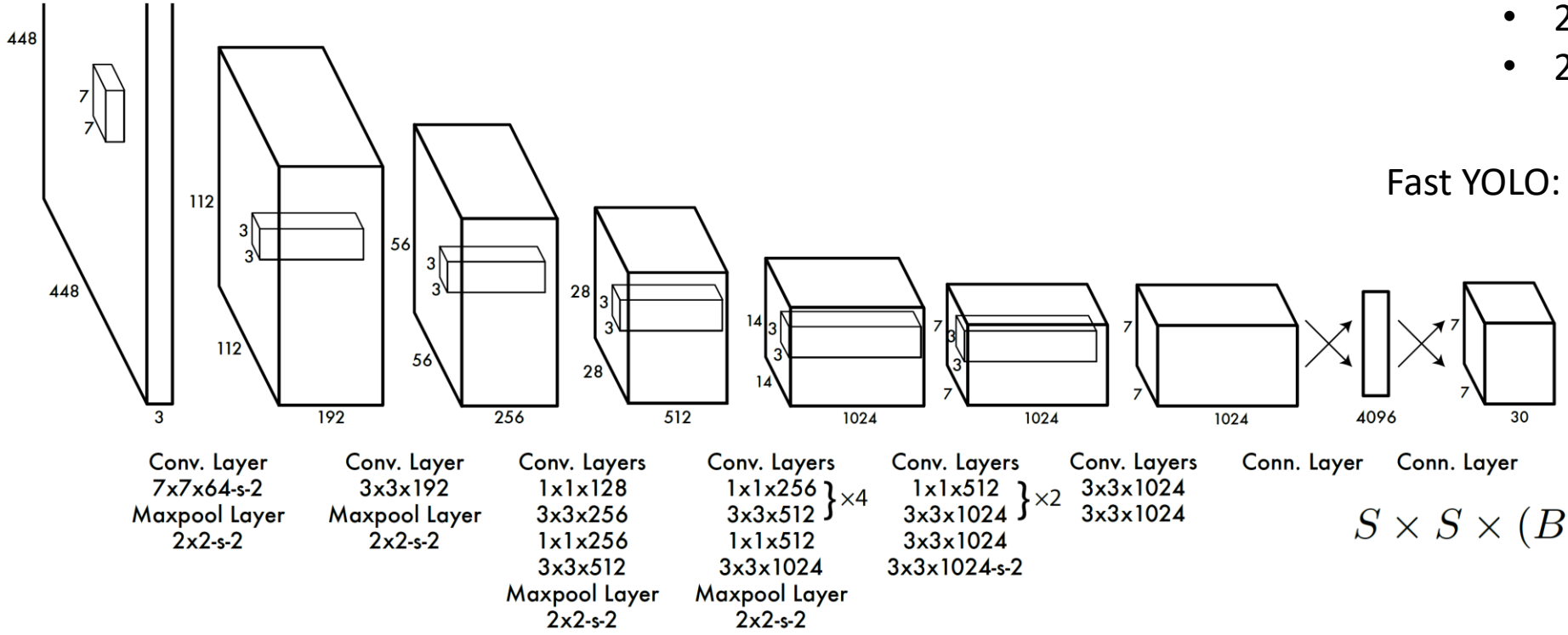
In testing, class-specific confidence scores for each box

$$\Pr(\text{Class}_i | \text{Object}) * \Pr(\text{Object}) * \text{IOU}_{\text{pred}}^{\text{truth}} = \Pr(\text{Class}_i) * \text{IOU}_{\text{pred}}^{\text{truth}}$$

You Only Look Once: Unified, Real-Time Object Detection. Redmon et al., CVPR, 2016

YOLO

- Regress to bounding box locations and class probabilities



- 24 convs
- 2 FCs

Fast YOLO: 9 convs, less filter

$$S \times S \times (B * 5 + C)$$

You Only Look Once: Unified, Real-Time Object Detection. Redmon et al., CVPR, 2016

YOLO

- Training loss function

$$\lambda_{\text{coord}} \sum_{i=0}^{S^2} \sum_{j=0}^{B} \mathbb{1}_{ij}^{\text{obj}} \left[(x_i - \hat{x}_i)^2 + (y_i - \hat{y}_i)^2 \right] \quad \text{(x, y)}$$

$$+ \lambda_{\text{coord}} \sum_{i=0}^{S^2} \sum_{j=0}^{B} \mathbb{1}_{ij}^{\text{obj}} \left[\left(\sqrt{w_i} - \sqrt{\hat{w}_i} \right)^2 + \left(\sqrt{h_i} - \sqrt{\hat{h}_i} \right)^2 \right] \quad \text{(w, h)}$$

$$+ \sum_{i=0}^{S^2} \sum_{j=0}^{B} \mathbb{1}_{ij}^{\text{obj}} \left(C_i - \hat{C}_i \right)^2$$

$$+ \lambda_{\text{noobj}} \sum_{i=0}^{S^2} \sum_{j=0}^{B} \mathbb{1}_{ij}^{\text{noobj}} \left(C_i - \hat{C}_i \right)^2 \quad \text{(confidence)}$$

$$+ \sum_{i=0}^{S^2} \mathbb{1}_{i}^{\text{obj}} \sum_{c \in \text{classes}} \left(p_i(c) - \hat{p}_i(c) \right)^2 \quad \text{(class)}$$

$\mathbb{1}_{ij}^{\text{obj}}$ jth bounding box from cell i "responsible" for the prediction

highest current IOU with the ground truth

$\mathbb{1}_{i}^{\text{obj}}$ If object appears in cell i

$\lambda_{\text{coord}} = 5 \qquad \lambda_{\text{noobj}} = .5$

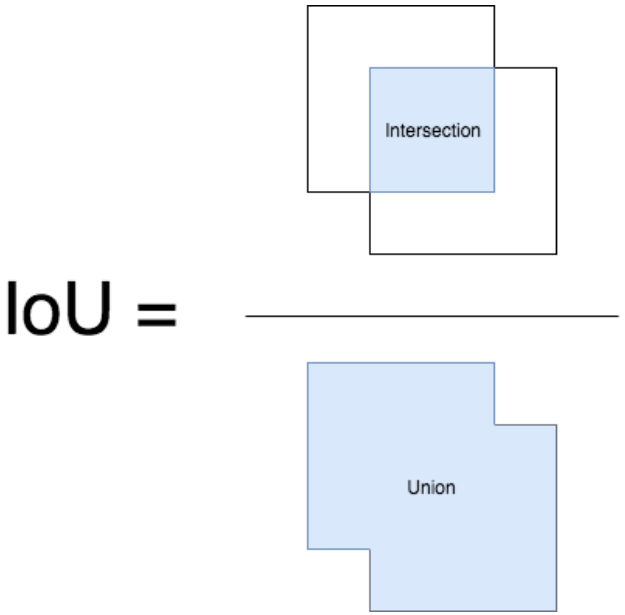
$p_i(c)$: predicted class probability

$\hat{p}_i(c)$: ground truth class (one-hot)

You Only Look Once: Unified, Real-Time Object Detection. Redmon et al., CVPR, 2016

Non-maximum Suppression

- Keep the box with the highest confidence/score
- Compute IoU between this box and other boxes
- Suppress boxes with IoU > threshold



https://towardsdatascience.com/non-maximum-suppression-nms-93ce178e177c

YOLO

Real-Time Detectors	Train	mAP	FPS
100Hz DPM [31]	2007	16.0	100
30Hz DPM [31]	2007	26.1	30
Fast YOLO	2007+2012	52.7	**155**
YOLO	2007+2012	**63.4**	45
Less Than Real-Time			
Fastest DPM [38]	2007	30.4	15
R-CNN Minus R [20]	2007	53.5	6
Fast R-CNN [14]	2007+2012	70.0	0.5
Faster R-CNN VGG-16[28]	2007+2012	73.2	7
Faster R-CNN ZF [28]	2007+2012	62.1	18
YOLO VGG-16	2007+2012	66.4	21

You Only Look Once: Unified, Real-Time Object Detection. Redmon et al., CVPR, 2016

YOLOv2 and YOLOv3

- YOLOv2
  - Batch normalization (normalization of the layers' inputs by re-centering and re-scaling)
  - High resolution classifier 416x416
  - Convolutional with anchor boxes (remove FC layers)
  - Dimension clustering to decide the anchor boxes
  - Bounding box regression
  - Multi-scale training (change input image size)
- YOLOv3
  - Binary cross-entropy loss for the class predictions
  - Prediction across scales

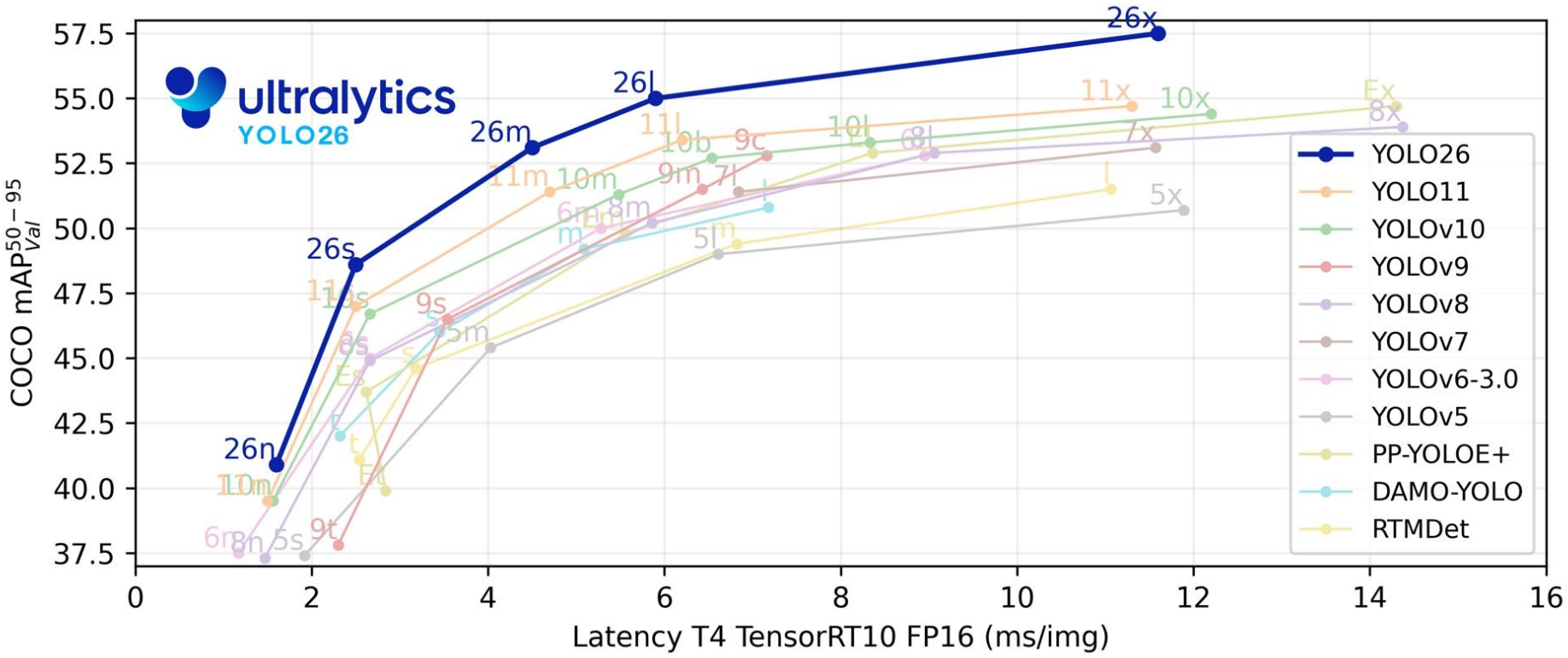
	Type	Filters	Size	Output
	Convolutional	32	3 × 3	256 × 256
	Convolutional	64	3 × 3 / 2	128 × 128
1×	Convolutional	32	1 × 1	
	Convolutional	64	3 × 3	
	Residual			128 × 128
	Convolutional	128	3 × 3 / 2	64 × 64
2×	Convolutional	64	1 × 1	
	Convolutional	128	3 × 3	
	Residual			64 × 64
	Convolutional	256	3 × 3 / 2	32 × 32
8×	Convolutional	128	1 × 1	
	Convolutional	256	3 × 3	
	Residual			32 × 32
	Convolutional	512	3 × 3 / 2	16 × 16
8×	Convolutional	256	1 × 1	
	Convolutional	512	3 × 3	
	Residual			16 × 16
	Convolutional	1024	3 × 3 / 2	8 × 8
4×	Convolutional	512	1 × 1	
	Convolutional	1024	3 × 3	
	Residual			8 × 8
	Avgpool		Global	
	Connected		1000	
	Softmax			

Table 1. **Darknet-53.**

YOLO9000: Better, Faster, Stronger. Redmon & Farhadi, CVPR, 2017
YOLOv3: An Incremental Improvement

Ultralytics YOLO

- https://docs.ultralytics.com/models/yolo26/



Average Precision in Object Detection

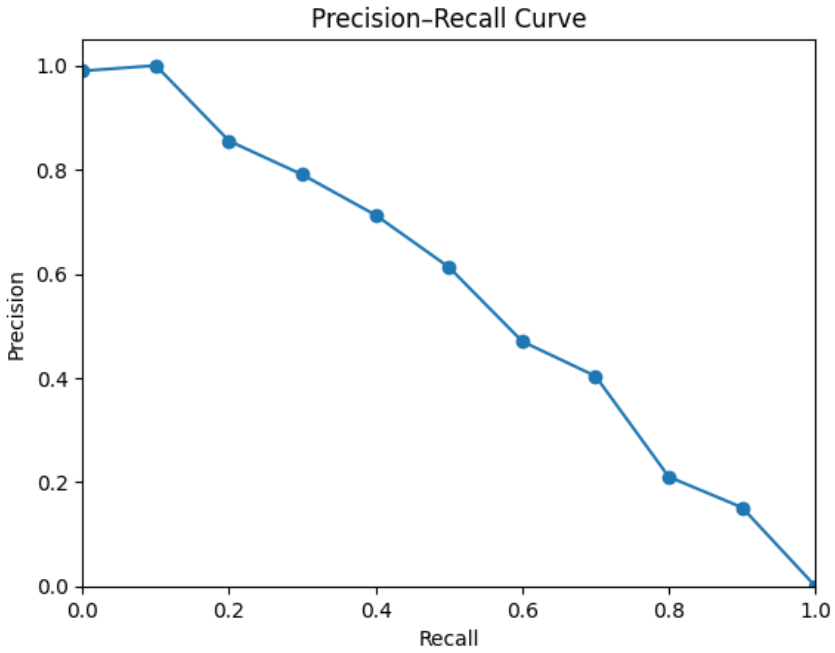
- Evaluates how good an object detector is
- Combines precision and recall
- Computed from the Precision–Recall curve

Intersection over Union (IoU)

IoU measures overlap between predicted and ground-truth boxes.
A detection is correct if IoU ≥ threshold (e.g., 0.5).

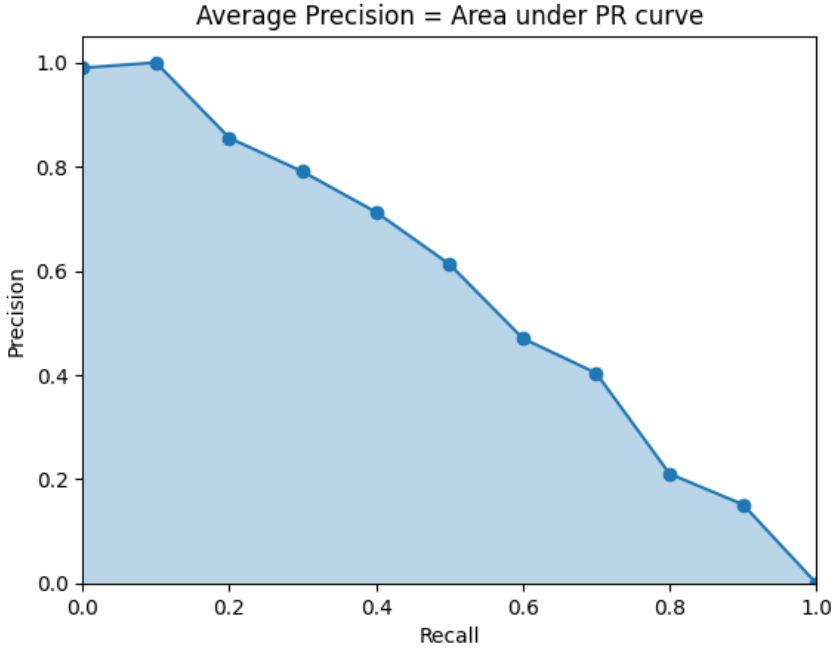
Precision–Recall Curve

Sort detections by confidence.
Gradually include predictions and compute precision and recall.
Plot precision vs recall.



Average Precision (AP)

AP = area under the Precision–Recall curve.
Higher AP means better detector performance.



Summary

- Two-stage detectors
  - R-CNN, Fast R-CNN, Faster R-CNN
  - Region proposal + classification
  - Good performance, slow

- One-stage detectors
  - YOLO, SSD
  - End-to-end network to regress to bounding boxes
  - Fast, comparable performance to two-stage detectors

Further Reading

- Viola–Jones object detection, 2001 https://www.cs.cmu.edu/~efros/courses/LBMV07/Papers/viola-cvpr-01.pdf
- Deformable part model, 2010, https://ieeexplore.ieee.org/document/5255236
- R-CNN, 2014 https://arxiv.org/abs/1311.2524
- Fast R-CNN, 2015 https://arxiv.org/abs/1504.08083
- Faster R-CNN, 2015 https://arxiv.org/abs/1506.01497
- YOLO, 2015 https://arxiv.org/abs/1506.02640
- YOLOv2, 2016 https://arxiv.org/abs/1612.08242
- Feature Pyramid Networks, 2017 https://arxiv.org/pdf/1612.03144.pdf
- DETR, 2020 https://arxiv.org/abs/2005.12872